Missing Values Analysis

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Packages

```
library(tidyverse)
library(misty)
library(lme4)
library(DescTools)
library(broom)
```

Data preparation

```
dat_aggregated <- read_csv("aggregated_data_triggered_ZI.csv")

dat_aggregated <- dat_aggregated %>%
    mutate(
      K1 = as.factor(K1),
      K2 = as.factor(K2)
    )
```

Missings summary

```
# Overall missings summary
summary_missings <- dat_aggregated %>%
    summarise(
      total = n(),
      completed = sum(interactionStatus == "Completed"),
      ignored = sum(interactionStatus == "Ignored"),
      incomplete = sum(interactionStatus == "Incomplete"),
```

```r
      completed_rate = mean(interactionStatus == "Completed"),
      ignored_rate = mean(interactionStatus == "Ignored"),
      incomplete_rate = mean(interactionStatus == "Incomplete")
    )

summary_missings
```

```
# A tibble: 1 x 7
  total completed ignored incomplete completed_rate ignored_rate incomplete_rate
  <int>     <int>   <int>      <int>          <dbl>        <dbl>           <dbl>
1 11685      5544    6051         90          0.474        0.518         0.00770
```

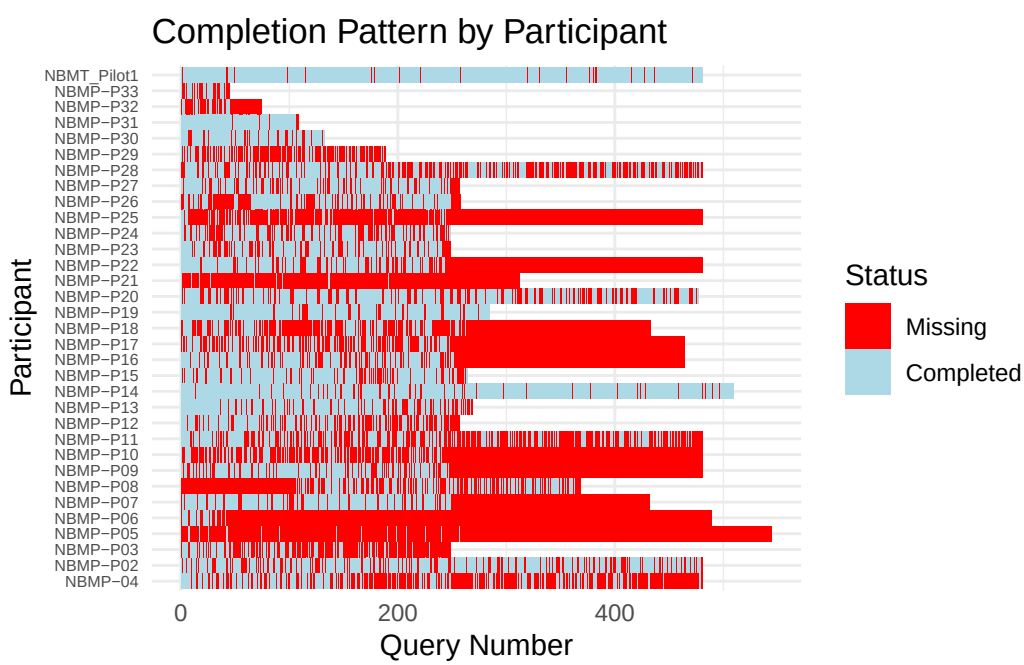
```r
# Participant summary
participant_missings <- dat_aggregated %>%
  group_by(pseudonym) %>%
  summarise(
    queries_total = n(),
    n_completed = sum(interactionStatus == "Completed"),
    n_ignored = sum(interactionStatus == "Ignored"),
    n_incomplete = sum(interactionStatus == "Incomplete"),
    completion_rate = mean(interactionStatus == "Completed"),
    study_duration_days = round(as.numeric(difftime(max(variableCreatedDateTime),
                                                    min(variableCreatedDateTime),
                                                    units = "days")), 0)
  ) %>%
  arrange(desc(completion_rate))

participant_missings
```

```
# A tibble: 33 x 7
   pseudonym   queries_total n_completed n_ignored n_incomplete completion_rate
   <chr>               <int>       <int>     <int>        <int>           <dbl>
 1 NBMT_Pilot1           480         460        15            5           0.958
 2 NBMP-P31              108         103         5            0           0.954
 3 NBMP-P14              509         467        37            5           0.917
 4 NBMP-P19              284         250        33            1           0.880
 5 NBMP-P30              132         108        22            2           0.818
 6 NBMP-P13              268         209        56            3           0.780
 7 NBMP-P15              264         203        59            2           0.769
 8 NBMP-P27              256         188        65            3           0.734
 9 NBMP-P02              480         349       125            6           0.727
```

```
10 NBMP-P23                248          177         64              7            0.714
# i 23 more rows
# i 1 more variable: study_duration_days <dbl>
```

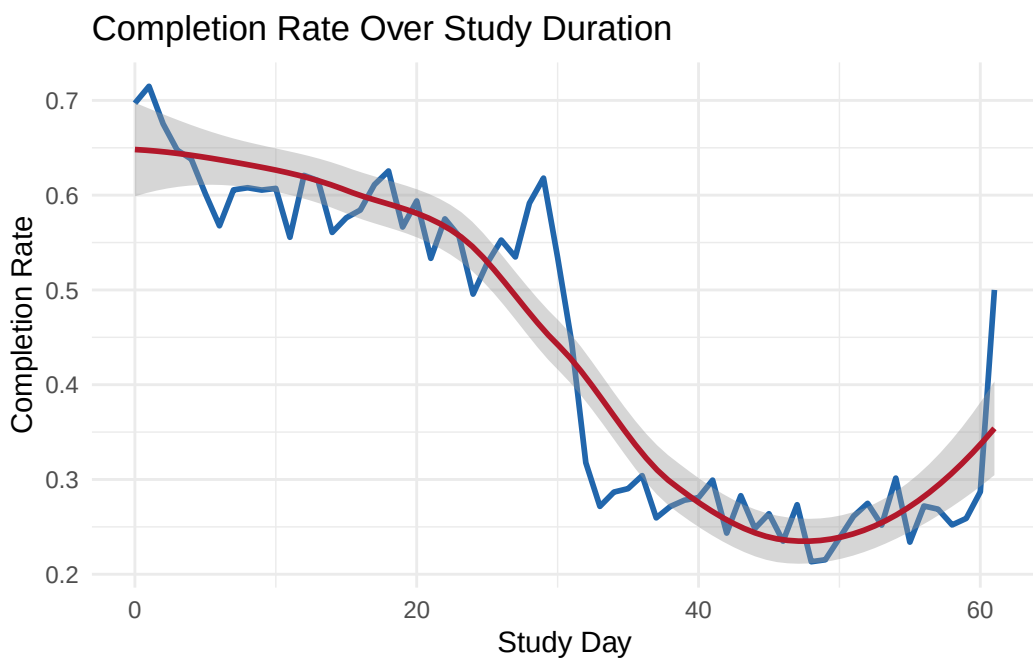
Heat map missings over time

```r
# Binary completion
dat_aggregated <- dat_aggregated %>%
  mutate(completed_binary = ifelse(interactionStatus == "Completed", 1, 0))

# Heatmap per participant
ggplot(dat_aggregated, aes(x = query_number, y = pseudonym, fill = factor(completed_binary)))
  geom_tile() +
  scale_fill_manual(values = c("0" = "red", "1" = "lightblue"),
                    labels = c("Missing", "Completed"),
                    name = "Status") +
  labs(title = "Completion Pattern by Participant",
       x = "Query Number",
       y = "Participant") +
  theme_minimal() +
  theme(axis.text.y = element_text(size = 6))
```



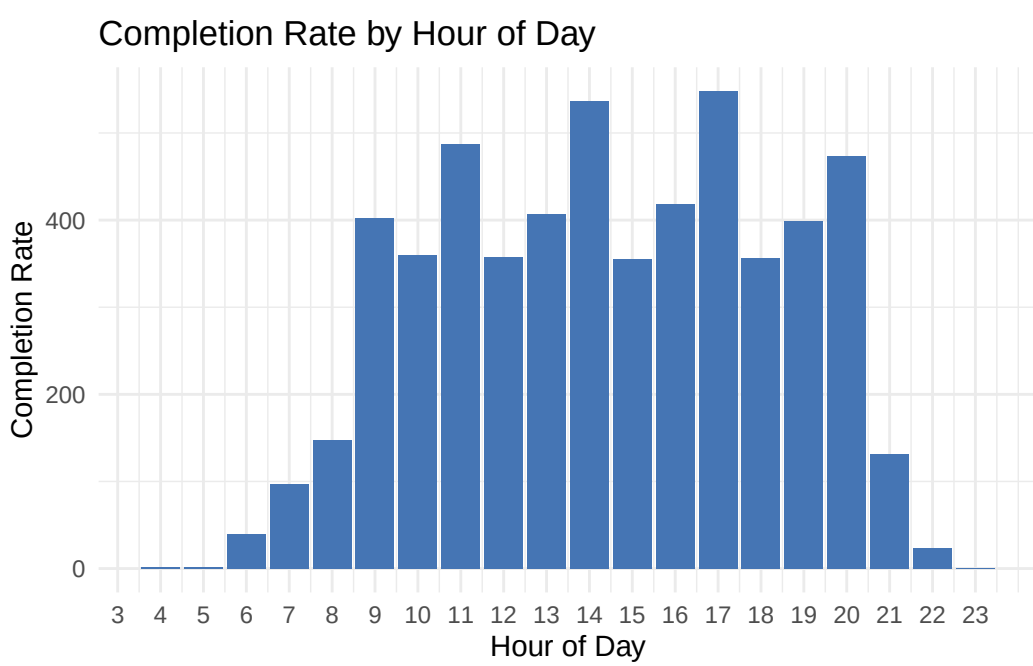
Plots completion trend, time of day, day of week

```
# Completion rate over study period
# Daily completion rate
daily_completion <- dat_aggregated %>%
  group_by(study_day) %>%
  summarise(
    completion_rate = mean(interactionStatus == "Completed"),
    n_queries = n()
  )

ggplot(daily_completion, aes(x = study_day, y = completion_rate)) +
  geom_line(color = "#2166ac", size = 1) +
  geom_smooth(method = "loess", se = TRUE, color = "#b2182b") +
  labs(title = "Completion Rate Over Study Duration",
       x = "Study Day",
       y = "Completion Rate") +
  theme_minimal()
```



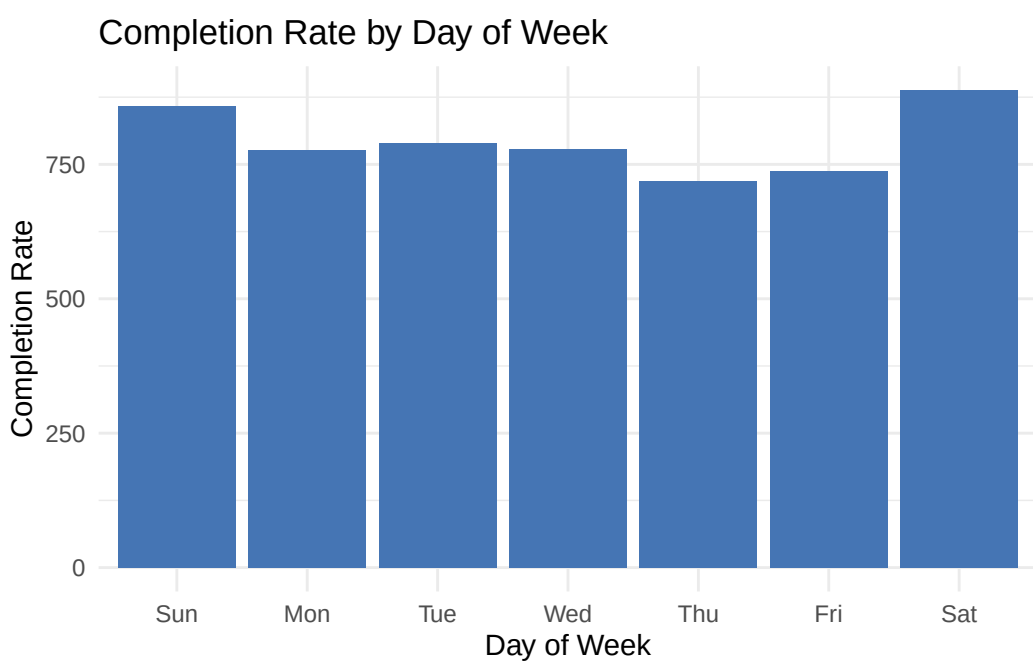

```
# Time of day patterns
hourly_completion <- dat_aggregated %>%
  mutate(hour = hour(variableCreatedDateTime))
```

```r
hourly_completion <- hourly_completion %>%
  group_by(hour) %>%
  summarise(
    completion_rate = sum(interactionStatus == "Completed"),
    n_queries = n()
  )

ggplot(hourly_completion, aes(x = hour, y = completion_rate)) +
  geom_col(fill = "#4575b4") +
  labs(title = "Completion Rate by Hour of Day",
       x = "Hour of Day",
       y = "Completion Rate") +
  scale_x_continuous(breaks = 0:23) +
  theme_minimal()
```




```r
# Weekday patterns
dat_aggregated <- dat_aggregated %>%
  mutate(weekday = wday(variableCreatedDateTime, label = TRUE))

weekday_completion <- dat_aggregated %>%
  group_by(weekday) %>%
  summarise(
```

```r
    completion_rate = sum(interactionStatus == "Completed"),
    n_queries = n()
  )

ggplot(weekday_completion, aes(x = weekday, y = completion_rate)) +
  geom_col(fill = "#4575b4") +
  labs(title = "Completion Rate by Day of Week",
       x = "Day of Week",
       y = "Completion Rate") +
  theme_minimal()
```



Test for trends, weekend effects, autocorrelation

```r
# Test temporal trend
#cor.test(daily_completion$study_day,
#         daily_completion$completion_rate)

# Test weekend differences
weekday_test <- dat_aggregated %>%
  mutate(weekend = weekday %in% c("Sat", "Sun"))
```

```r
t.test(completed_binary ~ weekend, data = weekday_test)
```

```
    Welch Two Sample t-test

data:  completed_binary by weekend
t = -5.3385, df = 6317.5, p-value = 9.698e-08
alternative hypothesis: true difference in means between group FALSE and group TRUE is not eq
95 percent confidence interval:
 -0.07421978 -0.03435164
sample estimates:
mean in group FALSE  mean in group TRUE
          0.4586403           0.5129260
```

```r
# Test autocorrelation in missings -> significant autocorr (but dropouts!)
dat_lag <- dat_aggregated %>%
  group_by(pseudonym) %>%
  arrange(variableCreatedDateTime) %>%
  mutate(
    prev_completed = lag(completed_binary)
  ) %>%
  ungroup()

table(dat_lag$completed_binary, dat_lag$prev_completed)
```

```
       0    1
  0 4684 1447
  1 1431 4090
```

```r
chisq.test(table(dat_lag$completed_binary, dat_lag$prev_completed))
```

```
    Pearson's Chi-squared test with Yates' continuity correction

data:  table(dat_lag$completed_binary, dat_lag$prev_completed)
X-squared = 2966.3, df = 1, p-value < 2.2e-16
```

```r
# Test autocorrelation in missings without dropouts
dat_lag_no_dropouts <- dat_lag %>%
  arrange(pseudonym, variableCreatedDateTime) %>%
  group_by(pseudonym) %>%
  mutate(
    valid_future = rev(cumany(rev(interactionStatus %in% c("Completed", "Incomplete"))))
  ) %>%
  filter(valid_future) %>%
  select(-valid_future) %>%
  ungroup()

table(dat_lag_no_dropouts$completed_binary, dat_lag_no_dropouts$prev_completed)
```

```
       0    1
  0 2031 1423
  1 1431 4090
```

```r
chisq.test(table(dat_lag_no_dropouts$completed_binary, dat_lag_no_dropouts$prev_completed))
```

```
    Pearson's Chi-squared test with Yates' continuity correction

data:  table(dat_lag_no_dropouts$completed_binary, dat_lag_no_dropouts$prev_completed)
X-squared = 968.19, df = 1, p-value < 2.2e-16
```

MNAR?

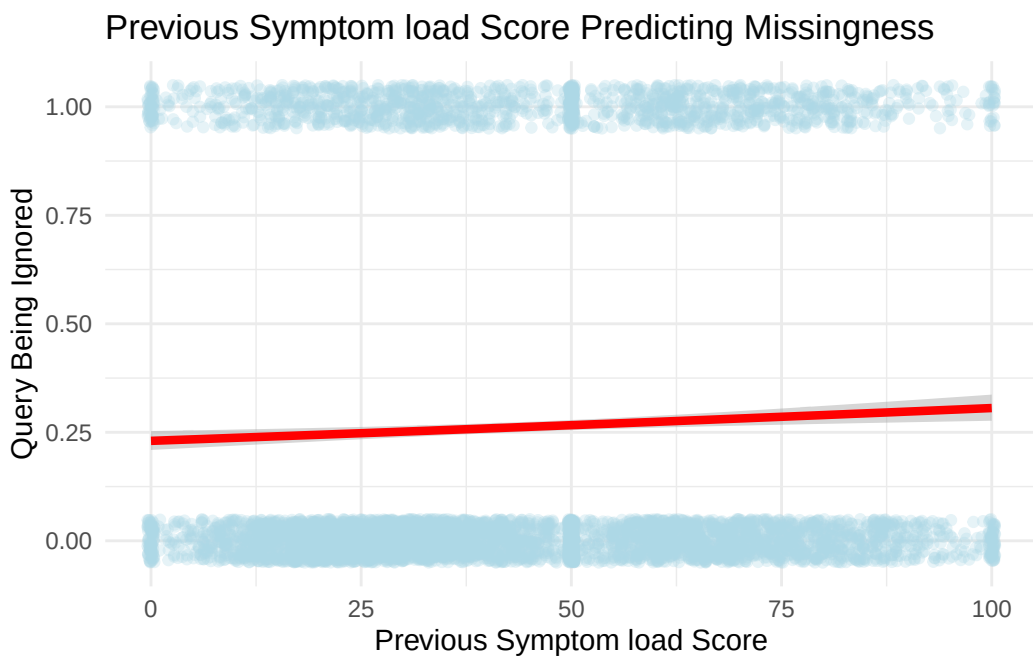
```r
# Lagged missings
dat_lag <- dat_lag %>%
  arrange(pseudonym, variableCreatedDateTime) %>%
  group_by(pseudonym) %>%
  mutate(
    # Ignored next
    next_ignored = lead(interactionStatus == "Ignored")
  ) %>%
  ungroup()

# Do previous scores predict next query being ignored?
```

```r
model_mood <- glmer(next_ignored ~ A1b + (1|pseudonym),
                    data = dat_lag,
                    family = binomial)
summary(model_mood)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: next_ignored ~ A1b + (1 | pseudonym)
   Data: dat_lag

      AIC       BIC    logLik -2*log(L)  df.resid
   5692.2    5712.1   -2843.1    5686.2      5687

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.0977 -0.5752 -0.4266  0.2483  5.2707

Random effects:
 Groups    Name        Variance Std.Dev.
 pseudonym (Intercept) 1.572    1.254
Number of obs: 5690, groups:  pseudonym, 33

Fixed effects:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.7509387  0.2331411  -3.221  0.00128 **
A1b         -0.0006853  0.0015262  -0.449  0.65340
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
    (Intr)
A1b -0.283
```

```r
# Visualization
ggplot(dat_lag, aes(x = A1b, y = as.numeric(next_ignored))) +
  geom_jitter(height = 0.05, alpha = 0.3, color = "lightblue") +
  geom_smooth(method = "glm", method.args = list(family = "binomial"),
              color = "red", size = 1.5) +
  labs(title = "Previous Symptom load Score Predicting Missingness",
       x = "Previous Symptom load Score",
```

```
       y = "Query Being Ignored") +
  theme_minimal()
```




```
dat_lag_c <- dat_lag %>%
  mutate(across(c("A1a", "A3", "A4", "A6", "A2", "A5", "A7"), scale))

# Test multiple preds
fit_missings_act <- glmer(next_ignored ~ A1b + A2 + A3 + A4 + A5 + A6 + A7 + (1|pseudonym),
                  data = dat_lag_c,
                  family = binomial)
summary(fit_missings_act)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: next_ignored ~ A1b + A2 + A3 + A4 + A5 + A6 + A7 + (1 | pseudonym)
   Data: dat_lag_c

      AIC       BIC    logLik -2*log(L)  df.resid
   5623.5    5683.3   -2802.8    5605.5      5620

Scaled residuals:
```

```
    Min      1Q  Median      3Q     Max 
-4.1981 -0.5694 -0.4193  0.2524  5.6839 

Random effects:
 Groups    Name        Variance Std.Dev.
 pseudonym (Intercept) 1.609    1.268   
Number of obs: 5629, groups:  pseudonym, 33

Fixed effects:
             Estimate Std. Error z value Pr(>|z|)   
(Intercept) -0.792203   0.244730  -3.237  0.00121 **
A1b          0.000324   0.002181   0.149  0.88192   
A2           0.075479   0.059297   1.273  0.20305   
A3           0.019507   0.059804   0.326  0.74429   
A4          -0.042621   0.053954  -0.790  0.42956   
A5           0.025633   0.057024   0.450  0.65306   
A6          -0.014831   0.050873  -0.292  0.77065   
A7           0.014311   0.053436   0.268  0.78885   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
    (Intr) A1b    A2     A3     A4     A5     A6    
A1b -0.381                                           
A2  -0.020  0.071                                    
A3  -0.158  0.397 -0.196                             
A4  -0.088  0.220 -0.180 -0.358                      
A5  -0.020  0.074 -0.250 -0.051 -0.052               
A6   0.002  0.030 -0.134  0.018 -0.023 -0.182       
A7  -0.001  0.002 -0.089 -0.050 -0.018 -0.265 -0.382
```

```r
fit_missings_A2 <- glmer(next_ignored ~ A1b + A3 + A4 + A5 + A6 + A7 + (1|pseudonym),
                     data = dat_lag_c,
                     family = binomial)
anova(fit_missings_act, fit_missings_A2, test = "LRT")
```

```
Data: dat_lag_c
Models:
fit_missings_A2: next_ignored ~ A1b + A3 + A4 + A5 + A6 + A7 + (1 | pseudonym)
fit_missings_act: next_ignored ~ A1b + A2 + A3 + A4 + A5 + A6 + A7 + (1 | pseudonym)
                 npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
fit_missings_A2     8 5623.2 5676.2 -2803.6    5607.2
```

```
fit_missings_act    9 5623.5 5683.3 -2802.8    5605.5 1.6169  1     0.2035
```

Effect of situation

```r
# differences in next missings depending on current situation
by_K1 <- dat_lag %>%
  group_by(K1) %>%
  summarise(next_miss = mean(next_ignored, na.rm = TRUE) * 100,
            n = n(), .groups = "drop") %>%
  mutate(next_miss = round(next_miss, 1))

by_K1
```

```
# A tibble: 4 x 3
  K1    next_miss     n
  <fct>     <dbl> <int>
1 1          25.7  3232
2 2          27.1  2266
3 3          23.5   171
4 <NA>       76.1  6016
```

```r
by_K2 <- dat_lag %>%
  group_by(K2) %>%
  summarise(next_miss = mean(next_ignored, na.rm = TRUE) * 100,
            n = n(), .groups = "drop") %>%
  mutate(next_miss = round(next_miss, 1))

by_K2
```

```
# A tibble: 13 x 3
   K2    next_miss     n
   <fct>     <dbl> <int>
 1 1          27.1   580
 2 2          28.8   341
 3 3          28.6   204
 4 4          26.2   713
 5 5          32.8   183
 6 6          32.2   323
 7 7          27.7   290
 8 8          29.1   489
```

```
 9 9          29.9    97
10 10         34.4   160
11 11         27.6   210
12 12         20.9   437
13 <NA>       64.5  7658
```

```r
# test linear effect of situation
fit_situation <- glm(next_ignored ~ K1 * K2,
                     data = dat_lag,
                     family = binomial)
summary(fit_situation)
```

```
Call:
glm(formula = next_ignored ~ K1 * K2, family = binomial, data = dat_lag)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.5368     0.1444  -3.718 0.000201 ***
K12          -0.7472     0.1964  -3.804 0.000142 ***
K13          -0.7625     0.4028  -1.893 0.058377 .
K22          -0.4475     0.2005  -2.232 0.025602 *
K23          -0.4907     0.2466  -1.990 0.046557 *
K24          -0.5721     0.1765  -3.241 0.001191 **
K25          -0.4600     0.2550  -1.804 0.071200 .
K26          -0.1701     0.2039  -0.834 0.404116
K27           0.7599     0.6862   1.107 0.268081
K28          -0.3861     0.1903  -2.029 0.042466 *
K29           0.1313     0.4787   0.274 0.783822
K210         -0.1563     0.2378  -0.657 0.510938
K211         -0.3795     0.2210  -1.717 0.085982 .
K212         -0.8381     0.2221  -3.774 0.000161 ***
K12:K22       1.0384     0.3400   3.054 0.002259 **
K13:K22      15.3129   535.4113   0.029 0.977183
K12:K23       1.0184     0.3774   2.699 0.006962 **
K13:K23       1.7900     1.4840   1.206 0.227735
K12:K24       1.0088     0.2745   3.675 0.000238 ***
K13:K24       0.8905     0.7943   1.121 0.262212
K12:K25       1.4127     0.3796   3.721 0.000198 ***
K13:K25      15.3254   535.4114   0.029 0.977165
K12:K26       0.5787     0.3264   1.773 0.076251 .
K13:K26       2.1626     1.2973   1.667 0.095520 .
```

```
K12:K27      -0.4720     0.7139  -0.661 0.508488
K13:K27      -0.5254     0.8641  -0.608 0.543209
K12:K28       0.9014     0.2875   3.136 0.001714 **
K13:K28     -11.8807   202.3668  -0.059 0.953184
K12:K29       0.1899     0.5592   0.340 0.734212
K13:K29     -12.3981   535.4115  -0.023 0.981526
K12:K210      0.9295     0.4557   2.040 0.041356 *
K13:K210      1.4556     1.4826   0.982 0.326181
K12:K211      0.3905     0.4994   0.782 0.434243
K13:K211      0.9856     1.3001   0.758 0.448383
K12:K212      0.7908     0.3117   2.537 0.011175 *
K13:K212      1.2619     0.6885   1.833 0.066845 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 4752.9  on 4019  degrees of freedom
Residual deviance: 4689.0  on 3984  degrees of freedom
  (7665 observations deleted due to missingness)
AIC: 4761

Number of Fisher Scoring iterations: 12
```

```r
# Pseudo R^2
PseudoR2(fit_situation, which="all")
```

```
       McFadden     McFaddenAdj        CoxSnell      Nagelkerke   AldrichNelson
   1.345826e-02   -1.690331e-03    1.578601e-02    2.276503e-02    1.566272e-02
VeallZimmermann           Efron McKelveyZavoina            Tjur             AIC
   2.891018e-02    1.566942e-02    1.287323e-01    1.566941e-02    4.760951e+03
            BIC          logLik         logLik0              G2
   4.987717e+03   -2.344476e+03   -2.376459e+03    6.396600e+01
```

```r
# LR test
fit_K2 <- glm(next_ignored ~ K2, data = dat_lag, family = binomial)
anova(fit_K2, fit_situation, test = "LRT")
```

```
Analysis of Deviance Table

Model 1: next_ignored ~ K2
```

```
Model 2: next_ignored ~ K1 * K2
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      4008     4731.4
2      3984     4689.0 24   42.419  0.01158 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
table_situation <- tidy(fit_situation, conf.int = TRUE) %>%
  filter(term != "(Intercept)") %>%
  arrange(desc(abs(estimate)))

table_situation
```

```
# A tibble: 35 x 7
   term     estimate std.error statistic  p.value conf.low conf.high
   <chr>       <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
 1 K13:K25     15.3    535.       0.0286 0.977     -93.7       NA
 2 K13:K22     15.3    535.       0.0286 0.977     -93.7       NA
 3 K13:K29    -12.4    535.      -0.0232 0.982      NA         96.6
 4 K13:K28    -11.9    202.      -0.0587 0.953      NA          6.03
 5 K13:K26      2.16     1.30     1.67   0.0955     -0.316      5.32
 6 K13:K23      1.79     1.48     1.21   0.228      -1.53       5.11
 7 K13:K210     1.46     1.48     0.982  0.326      -1.86       4.78
 8 K12:K25      1.41     0.380    3.72   0.000198    0.669      2.16
 9 K13:K212     1.26     0.689    1.83   0.0668     -0.137      2.60
10 K12:K22      1.04     0.340    3.05   0.00226     0.366      1.70
# i 25 more rows
```

Littles MCAR test

```r
# Littles MCAR test (but removes missings on all variables!!!) + large data set
na.test(dat_aggregated[,c("A1a", "A1b", "A3", "A9", "A4", "K3", "K4", "A8", "K1", "K2", "A6"
```

```
 Little's MCAR Test

     n nIncomp nPattern   chi2  df  pval
  5703    1693       18 485.65 137 0.000
```